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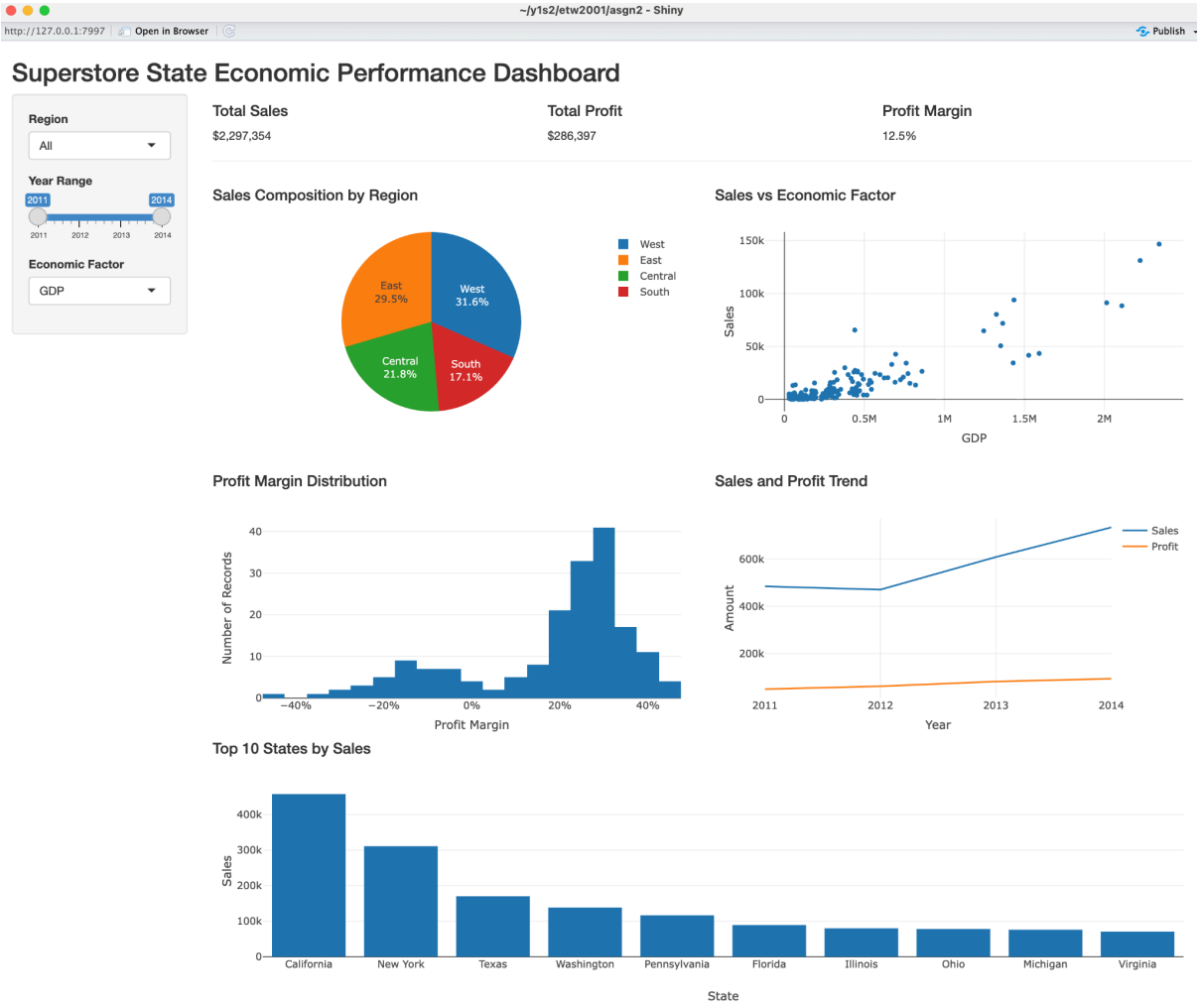


Figure 1.Superstore State Economic Performance Dashboard

Problem Statement:

This dashboard shows the superstore sales and profit performance for the USA and regions from 2011 to 2014. It includes three external data that GDP,population and income per capita.

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Data selection

Superstore dataset

The main dataset is used from assignment 1, it includes the information such as sales, profit, state,region and order date. In this databoard, the superstore dataset is used to analyse the sales performance, profit performance, profit margin, regional sales composition, yearly trends and the top performing states.

External dataset 1: GDP by State

Description of the dataset

The first dataset I choose is GDP by state, taken from the U.S. Bureau of Economic Analysis. This dataset provides the state annual gross domestic product (GDP) summary from 2011 to 2014, that matches the superstore years information.For example, California has the highest average GDP in the dataset, around 2,172,111 million dollars. In comparison, smaller states such as Vermont and Wyoming have much lower GDP values.

Justification

I chose this dataset because it can figure out if the superstore sales are related to the size of GDP or not. A state with higher GDP may have a better business profit and more active customer activity. For example, California has both the highest GDP and the highest Superstore sales in the combined dataset. The superstore sales show the internal business performance, while the GDP by state provides the external economy (U.S. Bureau of Economic Analysis, n.d.-a).

External Dataset 2: Income per Capita by State

Description of the dataset

The second dataset I choose is income per capita by state, taken from the U.S. Bureau of Economic Analysis. This dataset provides the income information per state, which is computed by dividing personal income by population. For example, Columbia has the highest average income per capita in the dataset, which is around $68,377. In comparison, states such as Mississippi, Idaho, New Mexico have lower average income per capita values (U.S. Bureau of Economic Analysis, n.d.-b).

Justification

I chose this dataset because it can reflect consumer purchasing power, the higher income per capita may prefer shopping in the superstore more than lower income state. It can explain the sales differences better because it shows the different levels of income across states.

External Dataset 3: Population by State

Description of the dataset

The third dataset I choose is population estimates for the USA from 2010 to 2019. I use the population information from 2011 to 2014 to match the Superstore years. The population data shows California has the largest average population in the dataset which is around 38.1 million people. In contrast, states such as Wyoming, Vermont and North Dakota have smaller populations (kbrookshier, n.d.).

Justification

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I chose this dataset because it represents the superstore market size, larger population may have more customers which can bring higher sales profit. For example, California has both the largest population and the highest superstore sales in the combined dataset, so the population is useful for figuring out the sales differences among those states.

Visualizations and Interpretation

Visualization 1 — Composition

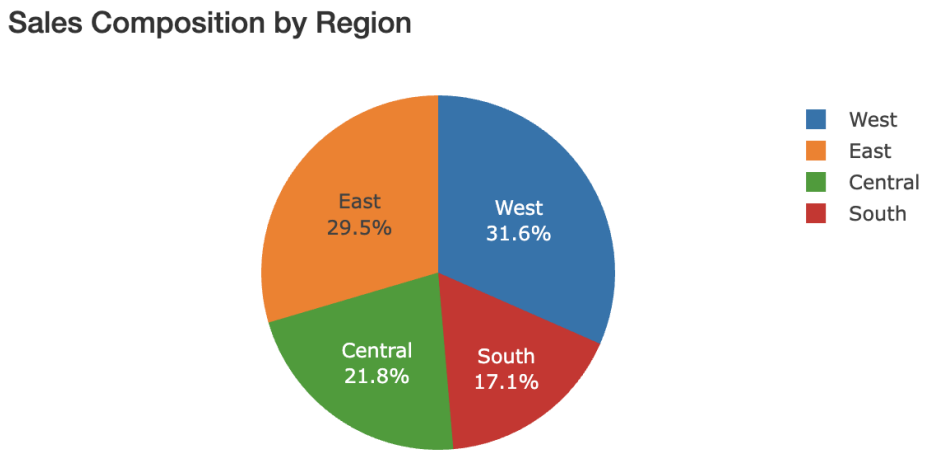


Figure 2. Sales Composition by Region

Business Question

How are the superstore sales distributed across different USA regions?

Chart Type

Pie chart, I use this chart to show the composition because it compare each region's share of total sales.

Interpretation

The pie chart shows that the west region contributes the largest share of total sales with around 31.6% of total superstore sales. The east region is contributing about 29.5%. Follow behind is the central region, which contributes about 21.8%, the last part is south region, with only around 17.1%. It shows the west and east regions are the two important sales regions that contribute more than half of the total superstore sales. The south region contributes the lowest sales during 2011 to 2014. So this chart shows which regions generate the largest and smallest parts of superstore sales to clearly overview the business performance.

Business Insight

This superstore should launch more products for strong regions such as west and east because they are the major sources of sales. At the same time, the company could check whether lower sales in the South are related to smaller market size, lower income to future optimize the product structure.

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Visualization 2 — Relationship

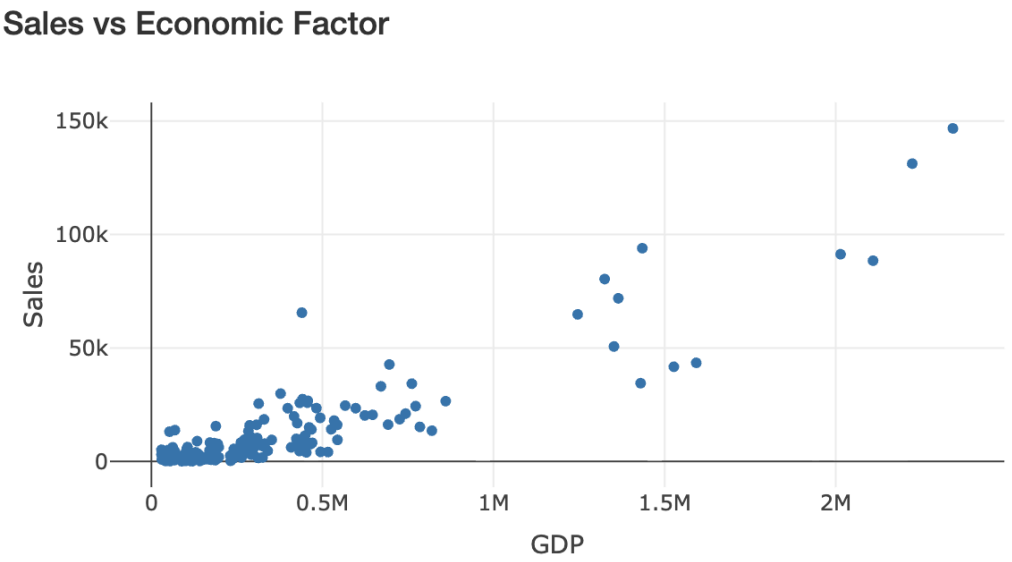


Figure 3. Sales vs Economic Factor: GDP Example

Business Question

Is the superstore's sales performance related to state economic size?

Chart Type

Scatter plot. I chose this because it can show the relationship between variables and Sales. In this picture, it shows the GDP in x-axis and show superstore sales in y-axis.

Interpretation

The scatter plot shows a positive relationship between GDP and superstore sales. Most states with lower GDP also have lower sales, while states with higher GDP tend to have higher sales. This suggests that superstore sales are earned mores in states with larger economies. For example, California has the highest average GDP in the dataset around $2172,111 million,and it also has the highest total Superstore sales which is around $457,731 from 2011 to 2014, it shows that economic size is related to superstore sales performance. However, the relationship is not perfect. For example, Texas has a higher average GDP than New York, but New York has higher Superstore sales. This means GDP is useful for understanding market potential, but it does not fully fit the sales performance by itself. This supports the purpose of combining the superstore dataset with external datasets.

Business Insight

The superstore should focus on the states with large economies because they may offer stronger sales opportunities. Such as California, New York, and Texas because they combine high sales with large economic size. At the same time, states such as Washington should also be studied carefully because they perform well even without being one of the largest GDP states.

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Visualization 3 — Distribution

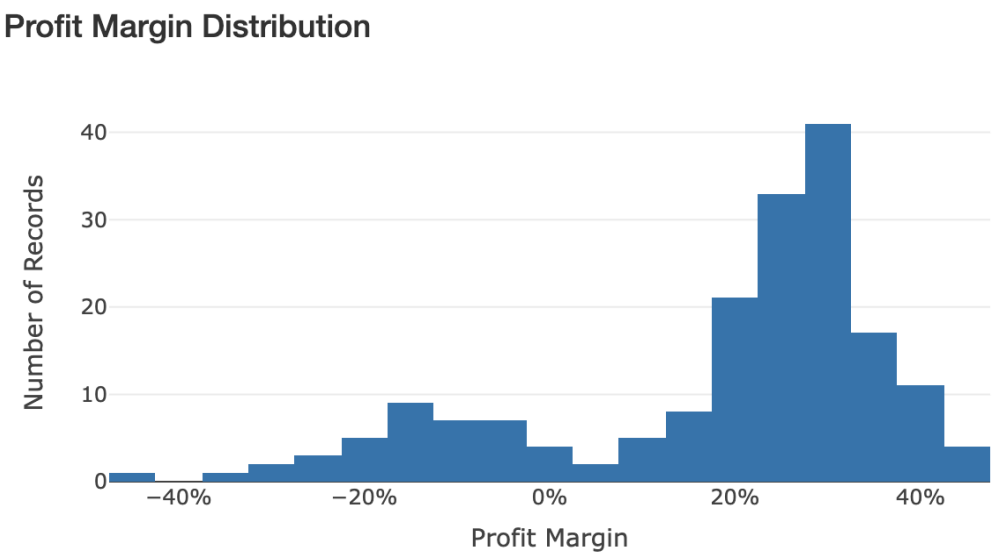


Figure 4. Profit Margin Distribution

Business Question

What is the distribution of superstore profit margins across state yearly records?

Chart Type

Histogram. I used this chart to show the distribution because it can show how often different profit margins appear in the dataset.

Interpretation

The histogram shows that most superstore states have positive profit margins. Based on the information, the profit margin ranges from about -45% to 45%, with a median profit margin of about 25%. This means that most records are profitable. A large number of records are concentrated around the 20% to 35% profit margin range. That means many states had reasonably profit performance. However, there are also 35 records with negative profit margins, which means some state records made a negative performance. For example, Colorado in 2014 had one of the lowest profit margins, around -45%.This shows that some areas may perform poorly for profit. This visualization shows a region may have sales, but the business also needs to check whether those sales are profitable. The distribution of profit margin helps identify whether poor performance is caused by low sales, low profitability, or both.

Business Insight

Superstore should investigate the state yearly records with negative profit margins. These losses may be related to high discounts, high shipping costs, or weak product profitability. The business should not only focus on increasing sales, but also improve profit margin by reviewing discounts. For lower performing states, superstore should make a good balance between the sales profit and promotional activities.

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Visualization 4 — Trends

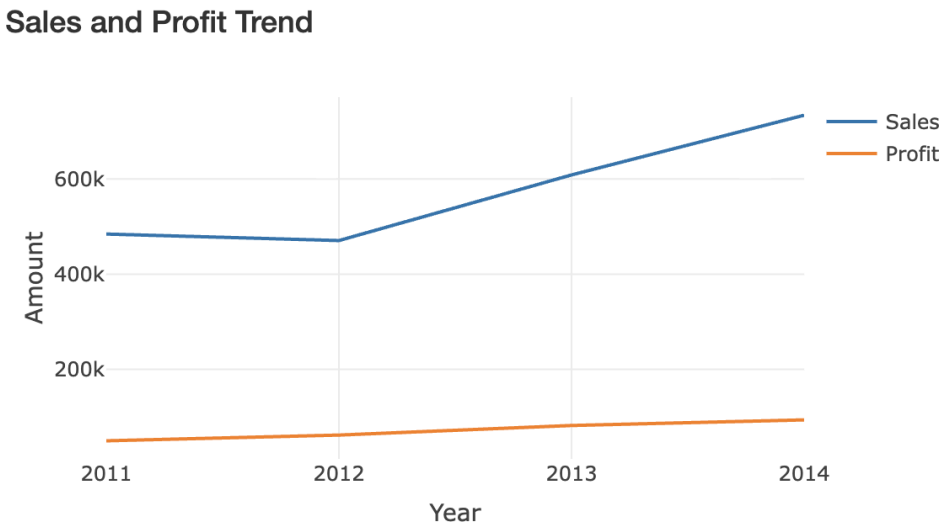


Figure 5. Sales and Profit Trend from 2011 to 2014

Business Question

How do superstore sales and profit change during 2011 to 2014?

Chart Type

Line chart. This chart can illustrate trends because it shows how sales and profit change across different years.

Interpretation

The line chart shows that superstore sales and profit increased from 2011 to 2014. Total sales were about $484,255 in 2011, slightly decreased to about $470,560 in 2012, then increased to about $734,016 in 2014.The profit followed an upward trend. It increased from about $49,544 in 2011 to $93,508 in 2014. This means that the business not only increased sales over time, but also improved profit performance. The trend shows that 2013 and 2014 were stronger years for superstore. This visualization helps answer the project problem by showing whether superstore performance improved or decreased over time, instead of only comparing areas. This is important because a region may look strong overall, but the yearly trend helps show whether the business is growing or not.

Business Insight

Superstore should figure out which factor contributed to the stronger performance in 2013 and 2014. If the increase was related to better sales strategy, or improved profit control, the company can continue using similar methods. The business should also investigate why sales slightly decreased in 2012 to avoid similar performance in the future.

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Visualization 5 — Comparison

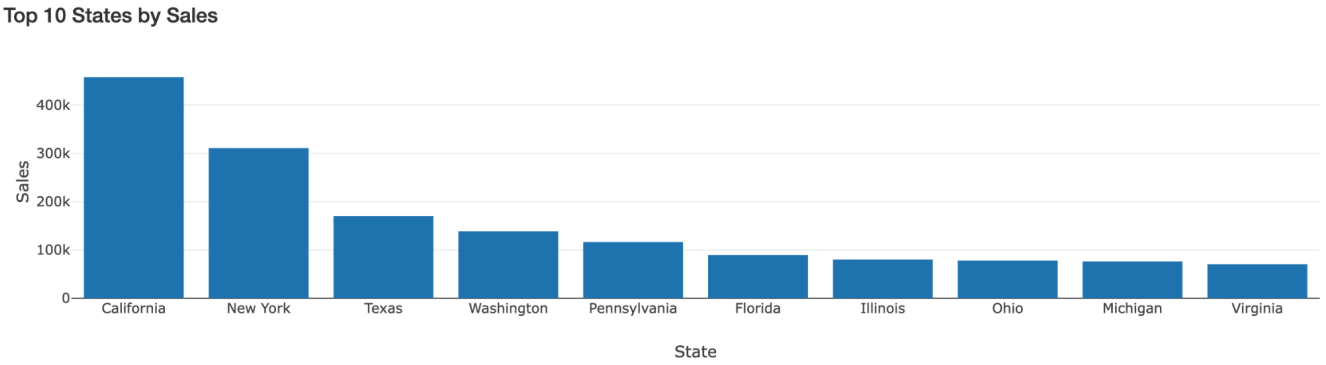


Figure 6. Top 10 States by Sales

Business Question

Which states generate the highest superstore sales?

Chart Type

Bar chart. This chart is used for comparison because it compares total sales across the top 10 states.

Interpretation

The bar chart shows that superstore sales are highly concentrated in a few states. California has the highest total sales, with about $457,731 from 2011 to 2014. New York is the second highest, with around $310,914. There is a large gap between California and the other states. California's sales are much higher than other states. This shows that California is the strongest state market in the dataset. However, high sales do not always mean high profit. For example, Texas is the third highest state by sales, but it has a negative total profit of about -$25,729. Pennsylvania, Florida, Illinois, and Ohio also appear in the top 10 sales states, but their total profits are negative. This means some states have strong sales but also have weak profit. This visualization helps identify the strongest state markets for superstore, because it shows exactly which states contribute the most sales.

Business Insight

Superstore should focus on higher sales states such as California and New York because they are major sources of profit. However, the company should not only rank states by sales. States such as Texas and Pennsylvania also have high sales but they owned negative profit. The business should check whether losses are caused by discounts, shipping costs, so that high sales states can become more profitable.

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Data Story

The dashboard shows that superstore performance from 2011 to 2014 was not balanced across regions, states and years. Overall, the business had strong sales and profit growth. By combining the superstore dataset with GDP, population, and income per capita data, the dashboard illustrated how business performance is related to state economic situations.

The first visualization shows the sales composition by region. Sales were mainly concentrated in the West and East regions. The West region contributed the largest share of sales, around 31.6%, followed by the East region, around 29.5%. In contrast, the South region had the smallest share, around 17.1%. That means that superstore rely on the West and East regions for sales. However, it also means the South region may need more attention because it contributes less to overall revenue. When choosing GDP as the external factor, the scatter plot shows that states with higher GDP generally have higher superstore sales. For example, California has both the highest GDP and the highest total sales in the dataset. New York and Texas also have high GDP and high sales. This suggests that larger state economies may provide more sales for superstore. However, sometimes GDP is disproportionate to superstore sales. For example, Texas has higher GDP than New York, but New York has higher superstore sales. This shows that business performance also depends on other factors.

The profit margin distribution gives another important part of the story. Most states for yearly records have positive profit margins, especially around the 20% to 35% range. This pointed out that many markets were profitable. However, some records had negative profit margins, with the lowest values below -40%. This means that sales alone cannot fully explain business sales. A state may generate sales, but if the profit margin is weak, the business may still face problems. Therefore, superstore should not only focus on revenue growth, but also pay attention to profitability.

The yearly trend shows that superstore performance improved over time. Sales decreased from about $484,255 in 2011 to $470,560 in 2012, but then increased strongly to about $608,523 in 2013 and $734,016 in 2014. Profit also increased from about $49,544 in 2011 to about $93,508 in 2014. This suggests that the business became stronger during these four year periods, especially in 2013 and 2014.

However, the state comparison shows that growth was not shared equally across all states. California was the strongest state market, with about $457,731 in total sales. New York was the second highest, with about $310,914. These states made a large contribution to overall superstore sales. This connects with the economic factor chart

because California and New York are also states with large economies or large populations, which may create stronger market opportunities.

At the same time, the dashboard shows that high sales do not always bring profit. For example, Texas had the third highest sales, but its total profit was negative. Pennsylvania, Florida, Illinois, and Ohio also appeared in the top 10 sales states, but they had negative total profit. This means superstore should not only focus on high sales, they also need to understand why some states generate strong revenue but earn poor profit.

Overall, the data story shows that superstore had positive growth from 2011 to 2014, and its sales were mainly concentrated in the West and East regions and the state is generally quite larger. External economic factors such as GDP, population, and income per capita help explain why some states may offer stronger market opportunities. However, profit margin analysis shows that sales growth alone is not enough. For future business decisions, superstore should protect strong markets such as California and New York, investigate high sales but low profit states such as Texas and Pennsylvania, and use economic factors to identify possible growth opportunities in other states.

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AI Acknowledgments

I acknowledge the use of ChatGPT (https://chat.openai.com/) to generate materials that were included within my final assessment in modified form. I entered the following prompts on 29 May 2026:

- Can you help me calculate the superstore total revenue and total profit by yearly?
- How can i combine the character with * and without * together cuz they are the same.
- Do i need to use ungroup there?